

M-containers: the typing of $\llbracket S, P \rrbracket_M$, and why it is your change-of-base, not a new theory

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for Neil Ghani

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This note answers your three messages (the THM 1 typing; “is that wise / what is its type”; “expand $M\text{-}\mathbf{Cont} \simeq \mathbf{Fam}(\mathbf{Kl}(M)^{\mathbf{op}})$ ”; functor vs. enriched functor; change-of-base vs. a separate $\llbracket - \rrbracket_M$ theory; the presheaf variant; and the headline). The short version: **you have caught a real fork, and you are right about which side of it is “just Set containers” — but the side I took is the other one, and it is exactly your change-of-base $\mathbf{Fam}(C^{\mathbf{op}})$ with $C = \mathbf{Kl}(M)$, not a new theory.**

1 The extension, exactly

For an M -container (S, P) , $P : S \rightarrow \mathbf{Set}$, my extension is

$$\llbracket S, P \rrbracket_M : \mathbf{Set} \rightarrow \mathbf{Set}, \quad \llbracket S, P \rrbracket_M(X) = \sum_{s:S} \mathbf{Kl}(M)(P_s, X) = \sum_{s:S} \mathbf{Set}(P_s, MX) = \sum_{s:S} (MX)^{P_s}.$$

Equivalently, and this is THM 1(b), *as ordinary $\mathbf{Set}$ -endofunctors and naturally in (S, P) ,*

$$\boxed{\llbracket S, P \rrbracket_M = \llbracket S, P \rrbracket \circ M}.$$

Its type is an ordinary endofunctor $\mathbf{Set} \rightarrow \mathbf{Set}$. The monad sits on the **values** X : a position $p \in P_s$, instead of holding a plain value, holds a Kleisli value $P_s \rightarrow MX$ (a P_s -indexed family of M -computations).

2 The fork you spotted

You wrote that you “expect the extension to be $\sum_{s:S} M(P_s) \rightarrow X$, which doesn’t seem to be what you have.” Correct — and the discrepancy is not a typo, it is a genuine design fork. There are two places to insert M :

	(i) M on values (mine)	(ii) M on positions (yours)
Formula	$\sum_s (P_s \rightarrow MX)$	$\sum_s (MP_s \rightarrow X)$
=	$\llbracket S, P \rrbracket \circ M$	$\llbracket (S, M \circ P) \rrbracket$
Lives in	$\mathbf{Fam}(\mathbf{Kl}(M)^{\mathbf{op}})$	ordinary $\mathbf{Cont} = \mathbf{Fam}(\mathbf{Set}^{\mathbf{op}})$
Effect on	the response / transition	the position <i>set</i>
New phenomena	codensity (THM 2), polynomiality (THM 3)	none — it is a Set container

Your instinct “these are all just Set containers” is exactly right *for (ii)*: $\sum_s (MP_s \rightarrow X) = \llbracket (S, M \circ P) \rrbracket$ is the ordinary extension of the container with fattened positions $M \circ P$. It is still a Set container; $\llbracket - \rrbracket$ on it is already full and faithful, and composition is ordinary **Cont** composition. There is nothing new to develop there — which is also why there is nothing new to *learn* there. All the content (probability keeps faithfulness; error keeps composability) lives on side (i), where the positions map into $\text{Kl}(M)$ rather than **Set**.

So: I have (i), not (ii); and (i) is chosen precisely because (ii) collapses back into plain containers.

3 Expanding $M\text{-Cont} = \text{Fam}(\text{Kl}(M)^{\text{op}})$

You asked me to expand this; it is an equality on the nose, not merely an equivalence.

- **Objects.** An object of $\text{Fam}(\text{Kl}(M)^{\text{op}})$ is a set S together with, for each $s \in S$, an object of $\text{Kl}(M)^{\text{op}}$. But objects of $\text{Kl}(M)$ (hence of $\text{Kl}(M)^{\text{op}}$) *are just sets*. So the data is S and a family $(P_s)_{s \in S}$ of sets — exactly a container.
- **Morphisms.** A morphism $(S, P) \rightarrow (S', P')$ in $\text{Fam}(D)$ is a function $f : S \rightarrow S'$ and, for each s , a D -morphism. For $D = \text{Kl}(M)^{\text{op}}$ a morphism $P_s \rightarrow P'_{f s}$ in $\text{Kl}(M)^{\text{op}}$ is a morphism $P'_{f s} \rightarrow P_s$ in $\text{Kl}(M)$, i.e. a function $\rho_s : P'_{f s} \rightarrow MP_s$. So a morphism is a shape map f plus, for each s , an *effectful response-translation* $\rho_s : P'_{f s} \rightarrow MP_s$ — exactly an M -container morphism.

The op is the usual container variance: positions travel backwards, now backwards *in* $\text{Kl}(M)$. The extension reads out through the covariant Kleisli representable $\text{Kl}(M)(P_s, -) = \mathbf{Set}(P_s, M-)$, giving the boxed formula above.

4 Functor, or enriched functor? — both, and the gap between them is the theorem

This is your sharpest question and it has real content. $\llbracket S, P \rrbracket_M$ is simultaneously:

- an **ordinary** functor $\mathbf{Set} \rightarrow \mathbf{Set}$ (target $[\mathbf{Set}, \mathbf{Set}]$), and
- the underlying functor of a $\text{Kl}(M)$ -**enriched** functor $\llbracket - \rrbracket^{\text{Kl}}$ (target $[\text{Kl}(M), \text{Kl}(M)]$), via $\llbracket - \rrbracket_M = R \circ \llbracket - \rrbracket^{\text{Kl}}$, where R restricts along the free functor $\mathbf{Set} \rightarrow \text{Kl}(M)$.

In the enriched reading $\llbracket - \rrbracket^{\text{Kl}}$ is *always* fully faithful — Kleisli-naturality is cheap. In the ordinary reading the honest target is $[\mathbf{Set}, \mathbf{Set}]$, and fullness can fail. The two targets differ by exactly the restriction $R =$ passing from Kleisli-naturality to pure-naturality, and **that gap is the codensity monad** $\Phi = \text{Ran}_M M$: by the Kan formula $\text{Nat}(\mathbf{Set}(A, M-), H) = (\text{Ran}_M H)(A)$, so $\llbracket - \rrbracket_M$ is full (on the honest target) iff M is codense and each $\mathbf{Set}(A, M-)$ is connected (THM 2). This is why probability (D codense) keeps faithfulness while Maybe/writer/reader lose it. (A note on your Lan_M instinct: the Kan structure is real but it is the *right* Kan extension Ran_M / codensity, not Lan ; that is the direction the adjunction $(-) \circ M \dashv \text{Ran}_M$ runs.)

5 This is your change-of-base, not a new theory

You said you are reluctant to treat $\llbracket - \rrbracket_M$ “as a whole different theory,” and that you like the K-container change-of-base framing ($C \rightarrow \mathbf{Set}$ semantics). **We agree, and $\llbracket - \rrbracket_M$ is that framing.**

It is not a parallel theory: it is $\text{Fam}(C^{\text{op}})$ — the generality you have already made priority #1 — specialised to $C = \text{Kl}(M)$:

$$M\text{-}\mathbf{Cont} = \text{Fam}(\text{Kl}(M)^{\text{op}}), \quad \mathbf{Cont} = \text{Fam}(\mathbf{Set}^{\text{op}}) \text{ is the case } M = \text{Id}.$$

The base change is on the *position category*, $\mathbf{Set} \xrightarrow{\text{free}} \text{Kl}(M)$; the ordinary $\mathbf{Set}$ -endofunctor $\llbracket - \rrbracket_M$ is the $\text{Fam}(C^{\text{op}})$ semantics read back along that base change, and codensity is precisely the obstruction to the readback being full. So the flagship $\text{Fam}(C^{\text{op}})$ theorem (T1: $\llbracket - \rrbracket$ fully faithful $\iff$ the monoidal unit is connected) governs the native ($\text{Kl}(M)$ -valued) semantics, and THM 2/THM 3 are what the instance $C = \text{Kl}(M)$ contributes on top: two independent Kan invariants of M — *codensity governs fullness, polynomiality governs composition*.

6 The presheaf / Cat variant

Your “shapes = small categories, positions = functors into $\mathbf{Set}$ ” is, I think, the *same* frame one base further out: $\text{Fam}(C^{\text{op}})$ with C a category of small categories (or $C = \mathbf{Cat}$), positions being presheaves/functors $P_s \rightarrow \mathbf{Set}$. This is exactly the Cat-case I sketched as future work for your UID-144 steer (`for-collaborator/2026-09-04-cont-future-work-cat-and-cont-cont.md`). It is a genuine instance, not a new theory, and it reinforces the point of §5: the unifying object is $\text{Fam}(C^{\text{op}})$, and each of $\{\mathbf{Set}, \text{Kl}(M), \mathbf{Cat}, \mathbf{Vec}\}$ is one choice of base. I am happy to develop the $\mathbf{Cat}$ instance next if you want it foregrounded rather than sketched.

7 Headline — my recommendation

You left the headline to me. Given the above I recommend: **frame the paper as change-of-base** ($\text{Fam}(C^{\text{op}})$, the case $C = \text{Kl}(M)$), and present codensity and polynomiality as the two theorems that this one instance yields. That keeps your preferred framing as the spine and makes the crown line — *two Kan invariants of the effect monad diverge oppositely* — a result inside it, not a rival theory. Title in that spirit: “Containers over a Kleisli base: codensity and polynomiality.”

What I would like from you

1. Confirm you are content with side (i) (effect on values, $\llbracket S, P \rrbracket \circ M$) as the object of study, with side (ii) noted as the “fattened Set container” degenerate case. If you would rather the effect sit on positions/response-selection, that is a modelling call I will follow — but it changes what is provable (side (ii) has no new fullness/composition phenomena).
2. Confirm the change-of-base spine + codensity/polynomiality-as-results framing for the write-up.
3. Say whether you want the $\mathbf{Cat}$ /presheaf instance developed now or left as a sketch.

With (1) settled you can “pick up” THM 2/3 as written — they are stated against side (i) exactly.